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1  # Note 05: Finding the Best Fitted Line (MSE Method)
2  # =====
3
4  # Import Libraries
5  import numpy as np
6  import matplotlib.pyplot as plt
7  import pandas as pd
8
9  # Load Dataset
10 df = pd.read_csv("test_data.csv")
11
12 # Define X and Y Variables
13 x = df['temperature']
14 y = df['voltage']
15
16 # Plot the Elbow Graph
17 orders = [1, 2, 3, 4, 5, 6]
18 MSE = []
19
20 # Calculate MAE Values
21 for order in orders:
22     # Fit polynomial
23     coefficient = np.polyfit(x, y, order)
24     # Create polynomial function
25     f = np.poly1d(coefficient)
26     # Predicted y values
27     yp = f(x)
28     # MAE calculation
29     mse = np.mean((y - yp)**2)
30     # Append Values
31     MSE.append(mse)
32
33 # Plot the Graphs
34 plt.figure()
35 plt.plot(orders, MSE, label="MSE", color="b")
36 plt.scatter(orders, MSE, label="MAE", marker="o", color="r")
37 plt.xlabel("Order")
38 plt.ylabel("MSE")
39 plt.title("MSE vs Order")
40 plt.grid(True, alpha=0.3)
41 plt.show()
42
43 # Plot the Smooth Curve
44 plt.figure()
45 plt.scatter(x, y, label="Original Data", marker="o", color="r")
46 x_new = np.linspace(min(x), max(x), 1000)
47 coefficient = np.polyfit(x, y, 4)
48 f = np.poly1d(coefficient)
49 yp = f(x_new)
50 plt.plot(x_new, yp, label="Best Fitted Line", color="g")
51 plt.xlabel("x")
52 plt.ylabel("y")
53 plt.title("Best Fitted Polynomial")
54 plt.grid(True, alpha=0.3)
55 plt.legend()
56 plt.show()
57
58 # Get Minimum MSE Order
59 print("Best Order : " + str(np.argmin(MSE[0:3])+1))

```